

Geospatial and Service-Level Analysis of NYC Benefits Access Centers Near CUNY Campuses

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Code Repository: with code, report paper, data information, & datasets:

https://github.com/DanShash/CUNY_access_data_project

Abstract

Urban accessibility to public services depends not only on geographic proximity but also on the types of services offered across locations. This project analyzes the spatial relationship between City University of New York (CUNY) campuses and NYC Benefits Access Centers using open public datasets. First, geospatial analysis measures the distance between campuses and the nearest service centers to evaluate accessibility. Second, machine learning techniques are used to predict borough classifications from geographic features. Finally, an extended analysis applies regular expression feature engineering, pivot-table aggregation, and visualization to explore borough-level differences in service characteristics. The results show that most CUNY campuses are geographically close to benefits centers, while service characteristics vary meaningfully across boroughs.

Research Questions

This project investigates three primary questions:

1. **Accessibility:**
Are NYC Benefits Access Centers geographically accessible to CUNY campuses?
2. **Predictive Spatial Structure:**
Can geographic features alone predict the borough classification of service centers?
3. **Service Characteristics:**
Do service offerings differ across boroughs even when geographic coverage exists?

Data

Two public datasets were used from [NYC open data](#) & [Data.gov](#) website sources.

CUNY Campus Locations

- 26 campuses across the five boroughs
- Features include campus name, latitude, and longitude

NYC Benefits Access Centers

- 29 locations after filtering administrative units
- Includes center name, borough, latitude, longitude, and descriptive text fields

Data was obtained from government open-data portals and processed using Python.

Tools and libraries used

- Python
- Pandas / NumPy
- Geopy
- Matplotlib / Seaborn
- Folium
- PyTorch
- Scikit-learn

Methodology

Data Preparation

Preprocessing steps included:

- Removing administrative or irrelevant columns
- Filtering data to the five official NYC boroughs
- Ensuring latitude and longitude values were numeric
- Standardizing column names across datasets

This produced a clean dataset suitable for geospatial analysis and modeling.

Geospatial Accessibility Analysis

To evaluate accessibility, the geodesic distance between each CUNY campus and all Benefits Access Centers was calculated using the **Geopy** library.

For each campus:

Distance{campus} = min{distance to all access centers}

This produced a dataset containing:

- campus coordinates
- nearest service center
- geographic distance between them

Key Findings

The distance analysis showed that:

- Most CUNY campuses are located **within 1–2 km** of a benefits access center
- Some campuses are farther from services, including **Kingsborough Community College**, which is more than **4 km** away

These results suggest that while geographic coverage is generally strong, certain areas may have limited physical access.

Predictive Modeling

To explore the predictive power of spatial data, a neural network was built to classify the borough of each access center.

Model Inputs

- Latitude
- Longitude
- Distance to nearest CUNY campus

Model Architecture

A small **Feedforward Neural Network (FNN)** implemented in PyTorch:

- Input layer: 3 neurons
- Hidden layer: 8 neurons (ReLU activation)
- Output layer: 5 neurons (borough classes)

Training details:

- Loss function: Cross-Entropy Loss
- Optimizer: Adam
- Training epochs: 100

Results

The model achieved approximately:

~75% classification accuracy

using only spatial features.

This result demonstrates that geographic coordinates and proximity to institutions contain meaningful signals for urban classification tasks.

Extended Analysis: Service Characteristics

The second phase of the project expanded beyond proximity analysis to examine **service characteristics across boroughs**.

Three techniques were applied:

1. **Regular Expression Feature Engineering**
2. **Pivot Table Aggregation**
3. **Visualization**

Feature Engineering with Regular Expressions

Unstructured text fields (such as service descriptions and comments) were processed using **regular expressions** to extract structured features.

The following binary features were created:

- presence of **child-related services**
- **specialized service centers**
- **family service or call center designation**

- **limited in-person service availability**

These engineered features allowed the dataset to be analyzed beyond raw text descriptions.

Borough-Level Analysis with Pivot Tables

Pivot tables were used to aggregate the engineered features at the borough level.

Key comparisons included:

- total number of access centers per borough
- prevalence of child-related services
- distribution of limited in-person services

Findings

Clear borough-level differences emerged:

- **Brooklyn:** All access centers provide child-related services and operate with limited in-person service models
- **Manhattan:** Similar service patterns with strong coverage
- **Bronx and Queens:** More mixed service availability
- **Staten Island:** Only one center, lacking the identified service features

This demonstrates that accessibility varies not only by distance but also by **service type and operational model**.

Visualization

Several visualizations were created to support the analysis:

Interactive Map (Folium)

Displays the geographic distribution of CUNY campuses and benefits centers across NYC.

Distance Analysis Charts

- bar charts showing distance between campuses and nearest centers
- histogram of campus-to-center distances

Borough-Level Service Charts

- total centers per borough
- grouped bar charts of service characteristics

These visualizations highlight both geographic coverage and differences in service characteristics across boroughs.

Conclusion

This project analyzed the accessibility of NYC Benefits Access Centers relative to CUNY campuses using open urban datasets and multiple analytical techniques.

Key findings include:

- Most campuses are located within **1–2 km** of a benefits access center, indicating strong geographic coverage.
- Some campuses are located farther away, suggesting potential areas for improved service placement.
- A neural network using spatial features achieved **75% classification accuracy**, demonstrating the predictive value of geographic data.
- Borough-level analysis revealed meaningful differences in **service characteristics**, even where geographic coverage exists.

Overall, the project demonstrates how combining **geospatial analysis, feature engineering, and machine learning** can provide deeper insight into urban service accessibility.

Tools and Environment

All analysis was conducted using **Python in Google Colab**, with outputs including:

- interactive geographic map
- data visualizations
- processed datasets
- trained machine learning models